PROBABILITY AND PROBABILITY DISTRIBUTIONS

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MSc Biostatistics 13571-875: Principles of Statistical Inference

Outline

1. Introduction to Probability
2. Probability and Bayes' Theory
3. Random Variables
4. Special Discrete Distributions
   - Bernoulli Distribution
   - Binomial Distribution
   - Hypergeometric Distribution
   - Poisson distribution
   - Negative Binomial Distribution
5. Special Continuous Distributions
   - Normal Distribution
6. Bivariate Distributions

Course Information

- Course convenor: Dr Innocent Maposa
- Email: imaposa@sun.ac.za
- Teaching Staff: Dr Alfred Musekiwa and Dr Merga Belina
- Teaching Assistant: Lovemore Mapahla lmapahla@gmail.com
- Prerequesites: Basic Statistics Theory; Calculus and Matrix Algebra
- Class time: 9:00hrs-13:00hrs with Tutorials after lunch

Prescribed Text Books

1. Casella and Berger (2002). Statistical Inference
2. Lehmann and Casella (1998). Theory of Point Estimation
3. Gelman, Carlin, Stern, Dunson, Vertari and Rubin (2013). Bayesian Data Analysis (Recommended Reading)

BIOSTAT 13571-875 Assessments and code of conduct

- TWO ASSIGNMENTS contributing 50% to final mark
  1. Assignment 1: Distrubution Theory, Principles of Data Reduction, and Point Estimation (20%). Due Date: 03/05/2022
  2. Assignment 2: Interval Estimation, Hypothesis Testing, Asymptotic Evaluations, Non-parametric procedures, Exact Inference and resampling methodology (30%). Due Date: 17/05/2022
- Examination (50%)

Code of conduct

- The course is run online and participants in the course should be panctual. Sessions will start promptly at 9am
- University misconduct rules and guidelines apply
- Homework and assignments, Tests and Examination are individual work so DO NOT share with friends

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What is Probability?

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3. Probability is the cornerstone of statistics! Why?

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1. Probability? Define...
2. Why probability? Determine the likelihood of an event....or quantifying uncertainty in a random phenomenon!
3. Probability is the cornerstone of statistics! Why?
4. Statistics (Field): The study of methods and theories used in data analysis obtained from (Random Processes). These methods underlie all processes involved in data collection, analysis, interpretation and dissemination.

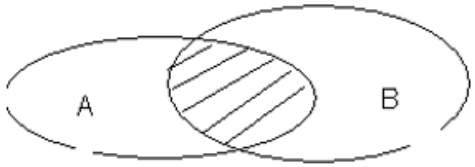
Some definitions...

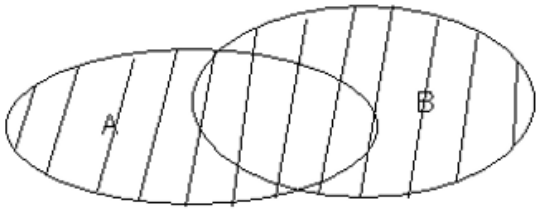
- Sample space: The set of all mutually exclusive and exhaustive possible outcomes of a random phenomenon or experiment.
- Event: Some description of the outcome of an experiment or random process...We seek to find the probability of such outcomes/events.
  - NB:Events of interest do not have to be mutually exclusive or exhaustive!
  - Terms that are used to describe the probability of an event include: chance, likelihood and probability!
- Exhaustive events: These are events whose Union, $\cup$, equals the sample space,S or Ω.
- Mutually exclusive events: These are events whose Intersection, $\cap$, has no elements, ie-events which have no common elements.
- An illustration of the ideas....

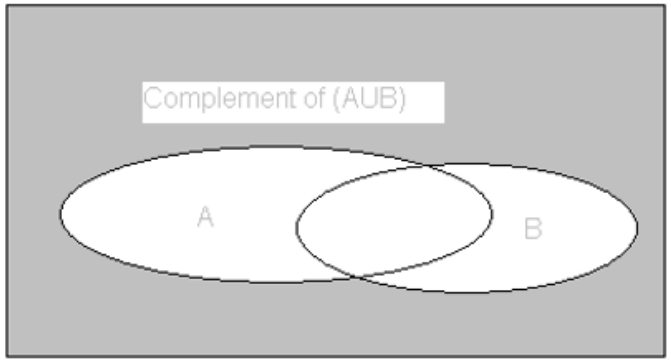
Operations with events

Operations on events

- Intersection
- Union
- Complement of an event







Operations on sets and events: Example-1

Let S, denote sample space, and A and B denote two events. Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{prime\ numbers\}$ and $B = \{even\ numbers\}$

- Represent this information in a diagram indicating the elements for $S, A\ and\ B$.
- Find:
  1. $A \cup B$ or ($A\ or\ B$)
  2. $A \cap B$ or ($A\ and\ B$)
  3. Are the events A and B exhaustive? Give reasons!
  4. Are the events A and B mutually exclusive? Give reasons!
- Define a sure event, an impossible event and a compliment for an event. ie-$P(S) =?$ and $P(S^c) =?$; $A^c =??$

Laws (Axioms) of Probability

- Formal definition (probability): A probability is a set function P that assigns to each event $A \in S$ a number $P(A)$; $[P : A \to P(A) \in [0, 1]]$ called the probability of the event A such that:
  1. $P(A) \geq 0$, that is, the probability of event A is non-negative *for all* $A \subseteq S$.
  2. The probability that some elementary event in S, the entire sample space will occur is 1: $P(S) = 1$.
  3. Any countable sequence of pairwise disjoint events $A_1, A_2, \ldots$ (that is, $A_i \cap A_j = \emptyset$ for all $i \neq j$) satisfies:

$$P(A_1 \cup A_2 \cup A_3 \ldots) = \sum_{\forall i} P(A_i)$$

- Ways of assigning probabilities includes:
  1. Classical method: n experimental outcomes, are equally likely: $Probability\ of\ each\ outcome = \frac{1}{n}$
  2. Relative frequency method: Event A occurs m times in n identical replications of an experiment: $P(A) = \frac{m}{n}\ as\ n \to \infty$
  3. Subjective method

Some key rules

- The additive rule: Let A and B be events defined in the sample space S, then:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

  and if the events are mutually exclusive $P(A \cup B) =?$
- The conditional probability rule: If A and B are events, then conditional probability of A occuring given B has occured, $(P(A|B))$, is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \forall \;\; P(B) \neq 0$$

- The independence rule: If A and B are events and they are independent, then $P(A \cap B) = P(A) * P(B)$
- An illustrative example:

Example 2: Basic probabilities

Table: Probability example: Patients visiting a clinic

Gender/Race	Female(F)	Male(M)	Total
Black(B)	65	50	-
White(W)	45	35	-
Colored(C)	30	45	-
Asian(A)	15	20	-
Total	-	-	-

Use information in table to find:

1. Probability of patient visiting clinic being female, $P(F) = ?$,
2. Probability of patient being white and male?
3. Probability of patient being Asian or Female?
4. Probability of patient being Female given the patient is Black?

Example 3: Disease diagnosis

Table: Probability example: Disease diagnosis

Diagnosis/Status	Present (D)	Absent (D^c)	Total
Positive (T^+)	22	51	-
Negative (T^-)	8	1739	-
Total	-	-	-

Use the information in this table to find:

1. Probability that a randomly chosen person has the disease, $P(D) =?$
2. Probability of testing positive given disease is present?
3. Probability of testing negative given disease is present?
4. Probability of disease absence given the test is positive?

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Bayes' Theory and some important rules

- The multiplicative rule: If A and B are events, then:

$$P(A \cap B) = \frac{P(A \cap B)}{P(B)} * P(B) = P(A|B) * P(B)$$

- The law of total probability: The law of total probability will help us use the multiplicative rule in more interesting ways. The law can be extended to many events but here we use 3 events to illustrate the rule:
  Suppose the sample space,S, is divided into 3 disjoint (mutually exclusive events), $A_1,\ A_2\ and\ A_3$, then for any event B in S (use diagram),

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3)$$

  This says that the probability of $"'B"'$ is the sum of it's pieces. This is equivalently written as:

Bayes' Theory and some important rules cont..

$$P(B) = P(B|A_1)*P(A_1)+P(B|A_2)*P(A_2)+P(B|A_3)*P(A_3) \quad (1)$$

This is the same as:

$$P(B) = \sum_{\forall i} P(B|A_i) * P(A_i) \quad (2)$$

In the multiplicative rule notation. Equation (1) or (2) known as:
The law of total probability
Consider the following example(s):

Bayes' Theory and some important rules cont..

1. A bag contains 5 white balls and 2 black balls. Two balls are drawn one after the other. What is the probability that the second ball is white? Use tree diagram to generate the probabilities for possible outcomes! Without conditioning first (with replacement) and with conditioning (without replacement)...
   - Define sample space: $S = \{WW, WB, BW, BB\}$
   - Define events: $W_1 = \{First\ ball\ is\ white\}$, $W_2 = \{Second\ ball\ is\ white\}$, $B_1 = \{First\ ball\ is\ black\}$ and $B_2 = \{second\ ball\ is\ black\}$.
   - Show that $P(W_2) = \frac{5}{7}$ using the law of total probability! First show using direct method...

Bayes' Theory and some important rules cont..

1. A bag contains 5 white balls and 2 black balls. Two balls are drawn one after the other. What is the probability that the second ball is white? Use tree diagram to generate the probabilities for possible outcomes! Without conditioning first (with replacement) and with conditioning (without replacement)...
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   - Show that $P(W_2) = \frac{5}{7}$ using the law of total probability! First show using direct method...
2. A bag contains 5 white balls and 2 blue balls. A ball is drawn. If it's blue a white ball is added to the bag and if it's white a blue ball is added to the bag. (Balls drawn without replacement). Then a second ball is drawn. What is the probability the second ball is white?

Bayes' Theory and some important rules cont..

- Following up on the Law of Total Probability we can state the Bayes' Theorem of probability and this theorem is fundamental in probability and statistics.
- The Bayes' Theorem (rule or formula) states that if A and B are two events, then:

$$P(B|A) = \frac{P(A|B) * P(B)}{P(A)} \tag{3}$$

Proof: Let A and B be 2 events, then:

$$P(B|A) * P(A) = P(B \cap A) = P(A|B) * P(B) \tag{4}$$

since $P(A \cap B) = P(B \cap A)$ and dividing both sides by $P(A)$ we get:

$$P(B|A) = \frac{P(A|B) * P(B)}{P(A)} \quad Bayes' \ formula \tag{5}$$

Bayes' Theory and some important rules cont..

- Equation (5) can be generalized using the law of total probability since P(A) can be written as:

$$P(A) = P(A \cap B) + P(A \cap B^c) \tag{6}$$

and for $B_1,\ B_2, \ldots,\ B_k$ mutually exclusive and exhaustive events, the Bayes' rule state that for any events $B_i:\ i = 1, \ldots, k$:

$$P(B_i|A) = \frac{P(A|B_i) * P(B_i)}{\sum_i P(A|B_i) * P(B_i)} \tag{7}$$

$$= \frac{P(A|B_i) * P(B_i)}{P(A|B_1) * P(B_1) + P(A|B_2) * P(B_2) + \cdots + P(A|B_k) * P(B_k)}$$

Bayes' Theorem Applications: Diagnosis

Consider the following events:

- D = "Disease is present"
- D^c = "Disease is absent"
- T^+ = "Positive test result (test detects disease)"
- T^- = "Negative test result (test does not detect disease)"

In diagnostic-testing situations, the following "performance parameters" of the diagnostic procedure under consideration will be available:

- $P(T^+|D)$ = "Sensitivity (true positive rate) of the test"
- $P(T^+|D^c)$ = "Probability of a false positive test result"
- $P(T^-|D)$ = "Probability of a false negative test result"
- $P(T^-|D^c)$ = "Specificity (or true-negative rate) of the test"

In addition, in order to derive estimates of the PVP (i.e., the predictive value of a negative test $PVN = P(D^c|T^-)$) we will need an estimate of the overall probability of disease in the general population. This is called the *prevalence* of the disease $P(D)$.

Goal: Find $P(D|T^+)$ the predictive value of a positive test result (or PVP), that is, find the probability that a subject has the disease given a positive test.

Now lets go back to example 3 and expand on the idea of finding probabilities when we know some other information external to our data.

Example 4: Bayes' formula application

- From Example 3; it is useful to find $P(D|T^+)$, that is, the probability of having disease given the diagnostic test is positive. This is also known as the predictive value of a positive test (PVP).
- Due to sampling bias, it may not be always correct to just estimate $P(D|T^+)$ based on observed data (in table). Thus we use the Bayes' theorem to find $P(D|T^+)$ and it says:

$$P(D|T^+) = \frac{P(T^+|D) * P(D)}{P(T^+)} \tag{8}$$

but $P(T^+) = P(T^+|D) * P(D) + P(T^+|D^c) * P(D^c)$-Law of Total Probabaility

- For example 3, define the probability of having disease before having data, ie-disease prevalence. This is known as the Prior probability. For our eg let $P(D) = 0.0093\%$.

Example 4: Bayes' formula application cont...

- Then we find the probability of testing positive given disease from the data. That is, the probability of data given likelihood (model). $P(T^+|D) = \frac{22}{30} = 0.7333$
- Thus Bayesian method is to make inference based on the posterior probability, calculated from the prior probability and the probability of the data given the likelihood (model) using the theorem. This concept will be extended for parameter estimation(s) given some probability density functions in Principles of Statistical Inference (Session 7).
- Going back to -eg-3: $P(T^+) = P(T^+|D)*P(D)+P(T^+|D^c)*P(D^c) = 0.02855717$
- Then $P(D|T^+) = \frac{P(T^+|D)*P(D)}{P(T^+)} = \frac{0.0000682}{0.02855717} = 0.002388$
- This implies the probability that an individual with a positive diagnosis has disease is about 26 times greater than the probability for an individual randomly chosen from the population. -ie-$\frac{P(D|T^+)}{P(D)} = 25.7$

Exercise 1

Q1 Let's consider one condition, this time of a combination of symptoms for chronic (ongoing) heart failure; [that is, Displaced apex (DA), JVD (jugular venous distension), gallop (G), rales[R] (crackling sound) and edema[E] (swelling) all physical findings seen in patients with heart failure]:

Table: CHF diagnosis

Diagnosis/Status	CHF	No CHF	Total
DA and others [T^+]	18	3	-
No DA or DA alone [T^-]	23	215	-
Total	-	-	-

Use the information in this table to find:

Exercise 1 cont...

1. Sensitivity; -ie-$P(T^+|CHF) = ?$
2. Specifity; -ie-$P(T^-|No\ CHF) = ?$
3. PVP (Predictive value of a Positive test) from the data; -ie-$P(CHF|T^+) = ?$
4. PVN (Predictive value of a Negative test) from the data; -ie-$P(CHF^c|T^-) = ?$

Q2. In a certain hospital 15% of the patients have got the HIV virus (D). Suppose a blood test is carried out on a patient. If the patient has got the virus the test will turn out positive (T^+) with probability 95%. If the patient does not have the virus the test will turn out positive with probability 2%. Find:

1. $P(D|T^+) = ?$
2. $P(D^c|T^+) = ?$
3. $P(D|T^-) = ?$
4. $P(D^c|T^-) = ?$

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Random variable: Definitions

- A random variable defined as a rule (ie function) that associates a number with each outcome in the sample space. In other words a random variable X is a function from the sample space Ω to the real numbers i.e to every possible outcome $\omega \in \Omega$, we have an associated real number $X(\omega)$.RVs are represented by capital letters eg X,Y etc
- A discrete random variable can take a countable and a whole number of values.
- A continuous random variable can take any value along a given interval of a number line. Mainly variables that are measurable.
- A probability distribution: The listing of all outcomes of an experiment or phenomenon and their associated probabilities. A probability distribution can be represented as a table, graph or a mathematical function.
- Probability distribution functions can either be discrete or

Discrete Probability Distribution Functions

- Example: Tossing a fair coin twice

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- NB: $\sum_{\forall x} P(X = x) = 1$

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- Define the random variable
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- NB: $\sum_{\forall x} P(X = x) = 1$
- Plot the function

Discrete Probability Distribution Functions

- Example: Tossing a fair coin twice
- Define the random variable $X = \{\textit{The number of heads after the two tosses}\}$.
- Generate a probability distribution function for X (table).
- NB: $\sum_{\forall x} P(X = x) = 1$
- Plot the function
- Exercise: Generate a probability distribution function for random variable Y defined as $Y = \{\textit{The number of heads after tossing a fair coin three times}\}$

Properties of probability distribution functions

- Mean: $\mu = E(X) = \sum_{\forall x} xP(X = x)$
- Variance: $\sigma^2 = E(X^2) - [E(X)]^2 = \sum_{\forall x} x^2 P(X = x) - \mu^2$
- Use the tossing coin example to find:
  1. E(X)
  2. Var(X)
- There are a number of special discrete probability distributions in statistics which includes Binomial, Poisson, Bernoulli, Geometric, Negative binomial, etc. In this session we look at a few of these...

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Bernoulli Distribution

- A random variable X has a bernoulli distribution if it can take only two values, that is, $X = 0$ *or* 1. With probability p and $1 - p$.
- Thus:

$$X = \begin{cases} 1 & \textit{if event A occured} \\ 0 & \textit{otherwise} \end{cases}$$

- The probability mass function is then represented as:

$$P(x) = \begin{cases} p^x(1-p)^{1-x} & x \in \{0, 1\} \\ 0 & \textit{otherwise} \end{cases}$$

- Some examples of situations where this distribution arises are: toss of a coin, success of a medical treatment, student passes exam etc.
- Let $X_1, X_2, \ldots, X_n$ be n bernoulli trials with parameter p, then the sum of the bernoulli independent events is a binomial r.v with parameter p. That is, if $Y = \sum_i^n X_i$, then Y are the total successes from n bernoulli trials!

Binomial Distribution

- The Binomial distribution is one of the most useful probability distributions in many experimental settings. -eg-tossing a coin, having disease and not having it, etc!
- The key components of a binomial distribution includes:
  1. n identical trials, -eg- 2 coin tosses
  2. 2 possible outcomes for each trial "'success"' and "'failure"', e.g. Head/Tail!
  3. Trials are independent, e.g. each coin toss doesn't affect the others
  4. $P("'success"') = p$, is the same for each trial -eg- $P(H) = 1/2$ for all trials....
- Normally represented as $X \sim Bin(n, p)$, where n is the number of trials and p is the probability of success.
- The functional form of the binomial distribution is:

$$P(X = x) = P(x) = \binom{n}{x} P^x (1-p)^{n-x} \qquad x = 0, 1, 2 \ldots, n \tag{9}$$

Example 5: Binomial probability

Suppose, we have an unfair coin for which the probability of getting a head is 2/3. Consider tossing the coin three times in a row and counting the number of times we observe the head. $X = \{No.\ of\ heads\ in\ 3\ coin\ tosses\}$ and $X = \{0, 1, 2, 3\}$. X is also a discrete R.V!

1. Find the probability that there are no heads after the three tosses.
2. Find the probability of getting at least 2 heads after the three tosses.
3. Find the probability of observing at most 1 head after the three tosses.

The mean and variance for $X \sim Bin(n, p)$ are given by the following expressions respectively:

- $E(X) = np$
- $Var(X) = npq$

Hypergeometric Distribution

- In the binomial situation, each trial was independent ie draw cards from a deck and replacing the drawn card each time.
- When the card drawn is not replaced, each trial now depends on the previous trial(s). The hypergeometric distribution is used in these situations.
- Suppose that a sample of size n is to be chosen randomly (without replacement) from an urn containing N balls, of which m are white and $N - m$ are black. If we let X denote the number of white balls selected, then
- Hypergeometric distribution:

$$P\{X = i\} = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}}; \quad i = 0, 1, \ldots, n$$

where N is the total number of elements, m are desired item from N elements, n are number of elements drawn and X are the number of successes in the n elements

Hypergeometric cont...

- The hypergeometric random variable X is the number of successes drawn from m available in the n selections.
- Example: A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are nondefective. If 30% of the lots have 4 defective components and 70% have only 1, what proportion of lots does the purchaser rejects?
- Solution: Let A denote the event that the purchaser accepts a lot. Now,

$$P(A) = P(A|lot\ has\ 4\ defectives)\frac{3}{10} + P(A|lot\ has\ 1\ defect)\frac{7}{10}$$

$$= \frac{\binom{4}{0}\binom{6}{3}}{\binom{10}{3}}\frac{3}{10} + \frac{\binom{1}{0}\binom{9}{3}}{\binom{10}{3}}\frac{7}{10} = \frac{54}{100}$$

Thus 46% of the lots are rejected.

The Poisson distribution

- The Poisson distribution is a discrete probability distribution for the counts of events that occur randomly in a given interval of time (or space)
- A R.V $X \sim Po(\lambda)$ when we get a discrete outcome which ranges from $X = \{0, 1, 2, \dots\}$.
- Experimental situations where we observe counts as outcomes are many and includes:
  1. Number of cases of a disease in different cities
  2. Number of births per hour during a given day at some clinic.
  3. Number of accidents per day at some place (-eg- busy junction)
  4. Number of domestic violence cases reported by couples in a survey

Poisson distribution cont...

- If $X = \{\textit{The number of events in a given interval}\}$, and the mean number of events per interval is λ, then the probability of observing x events in a given interval is:

$$P(X = x) = P(x) = \frac{\lambda^x e^{-\lambda}}{x!} \qquad x = 0, 1, 2, \ldots \qquad (10)$$

- Note that e is a mathematical constant, $e \approx 2.718282$ and λ is the parameter of the distribution.
- Example: births in a clinic occur randomly at an average rate of 1.5 births per hour.
  1. What is the probability of observing 4 births in a given hour at the hospital?
     $$P(X = 4) = ?$$
  2. What is the probability of observing at least 2 births in a given hour at the clinic?
- If $X \sim Po(\lambda)$, then $E(X) = \lambda$ and $var(X) = \lambda$

Other related distribution(s): Reflection

- Bernoulli and Binomial distributions: number of successes in independent trials (sampling with replacement) with fixed sample size.
- Hypergeometric distribution: number of successes in dependent trials (sampling without replacement) with fixed sample size.
- Poisson distribution: number of successes (events) occuring in a fixed interval of time and/or space without a fixed sample size
- In some cases, interest is on sample size necessary to get a certain number of successes.
  1. Geometric distribution (a special case of negative binomial): number of trials until the 1st success (including the success)
  2. Negative Binomial (NB) distribution: number of trials until the rth success (including the rth success).
  3. In both the Geometric and NB distributions, the trials are independent.

Negative Binomial Distribution

- Let X be a negative binomial distribution random variable ie the number of trials it takes to get the rth success.
- Parameters for the negative binomial are r and p where r are the number of successes required and p the probability of success in a single trial. $X \sim NB(r, p)$.

NB Distribution

$$P_X(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}; \quad x = r, r+1, r+2, \ldots$$

with $E(X) = \frac{r}{p}$ and $Var(X) = \frac{r(1-p)}{p^2}$

NB: Example

Innocent is required to sell chocolate bars to raise money for a trip. There is a 40% chance of him selling a chocolate bar at each potential customer. He has to sell 5 chocolate bars in all. Let X be the number of potential customers it takes to sell the required chocolate bars.

- What is r and p?
- What is the probability he sells his last chocolate bar at the 11th potential customer?
- What is the probability of Innocent finishing on or before the 8th potential customer?

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Normal Distribution

- Bell-shaped curve (also known as the Gaussian Distribution)
- Many random phenomena in the world can be modelled by a normal distribution and because of this, it's used as a basis for statistical inference
- PDF is fully determined by its mean and standard deviation ie $X \sim N(\mu, \sigma^2)$. Its also symmetric about its mean

Normal Distribution

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; \quad -\infty < x < \infty$$

for finite μ and $\sigma^2 < \infty$

- The CDF is defined as:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

Standard Normal Distribution

- The integral for the normal distribution is intractable as it is an infinite series
- The stanndard normal distribution, a special case of the normal distribution is well defined and used in determining any probability of interest for a normally distributed random variable.
- Standard Normal Distribution $Z \sim N(0,1)$ with PDF $\phi(z) = f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}; \quad -\infty < z < \infty$ and $\Phi(t) = F(t) = \int_{-\infty}^{t} \phi(z)dz$
- Any normally distributed random variable can be standardized (transformed) to a standard normal distribution and probabilities of interest generated
- NB: $P(a < Z < b) = \Phi(b) - \Phi(a)$

Normal Distribution: Example

Let X be IQ test scores known to be distributed with a mean of 100 and standard deviation of 15. If a random person takes this IQ test:

1. What is the probability that the IQ test score will be below 75?
2. What is the probability that the IQ test score will be more than 100?
3. What is the probability that the IQ test score will be between 85 and 115?

Other Distributions

- Unifrom distribution: $X \sim U(a, b)$ with pdf $f(x) = \frac{1}{b-a}; \quad a < x < b$
- Exponential distribution: $X \sim Exp(\lambda)$ with pdf $f(x) = \lambda e^{-\lambda x}; \quad x \geq 0$
- Gamma distribution: $X \sim Gamma(p, \lambda)$ with pdf $f(x) = \frac{\lambda^p}{\Gamma(p)} x^{p-1} e^{-\lambda x}; \;\; x \geq 0, \; \lambda > 0, \; p > 0$. This reduces to an exponential distribution when $p = 1$. Gamma function $\Gamma(p) = \int_0^\infty t^{p-1} e^{-t} dt$ and $\Gamma(p) = (p-1)!$
- Beta distribution: $X \sim Beta(\alpha, \beta)$ with pdf $f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)+\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}; \;\; x \in (0, 1)$

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Bivariate Distributions

- Two discrete and two continuous variables
- Consider two random variables X *and* Y whose values are $r,$ *and* s respectively and suppose that the probability of the event $X = r \cap Y = s$ is given by:

$$P(X = r, Y = s) = \begin{cases} \frac{r+s}{48}, & if 0 \leq r, s \leq 3 \\ 0, & otherwise \end{cases}$$

- create table for $X = r,$ *and* $Y = s$ and find different probabilities, marginals and conditional probability mass functions
- When there are two continuous random variables, the equivalent of the two-dimenional array is a region of the $x - y$ cartesian plane. Above the plane, over the region of interest is a surface which represents the probability density function associated with a bivariate distribution.

Bivariate continuous

- Suppose X and Y are two continuous random variables and that their values, *x and y* respectively, are constrained to lie within some region R of the cartesian plane.
- The associated pdf has the general form $f_{XY}(x, y)$ and as a function defining a surface over region R

$$\int\int_R f_{XY}(x, y)dxdy = 1$$

- Suppose X and Y are two continuous RVs with *x and y* constrained in the square $0 \leq x, y \leq 1$ and has pdf

$$f_{XY}(x, y) = \begin{cases} x + y, & \textit{if } 0 \leq x, y \leq 1 \\ 0, & \textit{elsewhere} \end{cases}$$

- a)Show that function is a pdf b)Find marginal of X, c) Find $P(0 \leq x, y \leq 0.5)$